

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
library(tidyverse)
library(rvest)
library(lubridate)
getwd()
```

```
## [1] "C:/Users/ /EDE_Spring2026"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Set the year to 2024
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
url_2024_durham <- "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024"
page_2024_durham <- read_html(url_2024_durham)
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
- Water system name
- Contact Person
- Ownership
- From the “3. Water Supply Sources” section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “Reginald Hicks”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3
the_website <- read_html(url_2024_durham)

tables <- the_website %>% html_elements("table") %>% html_table(fill = TRUE)

sys_table <- tables[[2]]

water_system_name <- as.character(sys_table[1, 2])
contact_person <- as.character(sys_table[4, 2])
ownership <- as.character(sys_table[2, 5])

water_system_name <- trimws(water_system_name)
contact_person <- trimws(contact_person)
ownership <- trimws(ownership)

MW_ID <- 10
mw_table <- tables[[13]]

mw_mat <- as.matrix(mw_table)

max_day_use_mgd <- c(
  mw_mat[1:4, 3],
  mw_mat[1:4, 6],
  mw_mat[1:4, 9]
)

max_day_use_mgd <- as.character(max_day_use_mgd)
max_day_use_mgd <- trimws(max_day_use_mgd)

water_system_name
```

```
## [1] "Durham"
```

```
contact_person
```

```
## [1] "Reginald Hicks"
```

```
ownership
```

```
## [1] "Municipality"
```

```
max_day_use_mgd
```

```
## [1] "34.50" "32.10" "38.20" "36.73" "36.06" "46.65" "41.90" "42.96" "37.33"
```

```
## [10] "37.36" "36.58" "34.45"
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

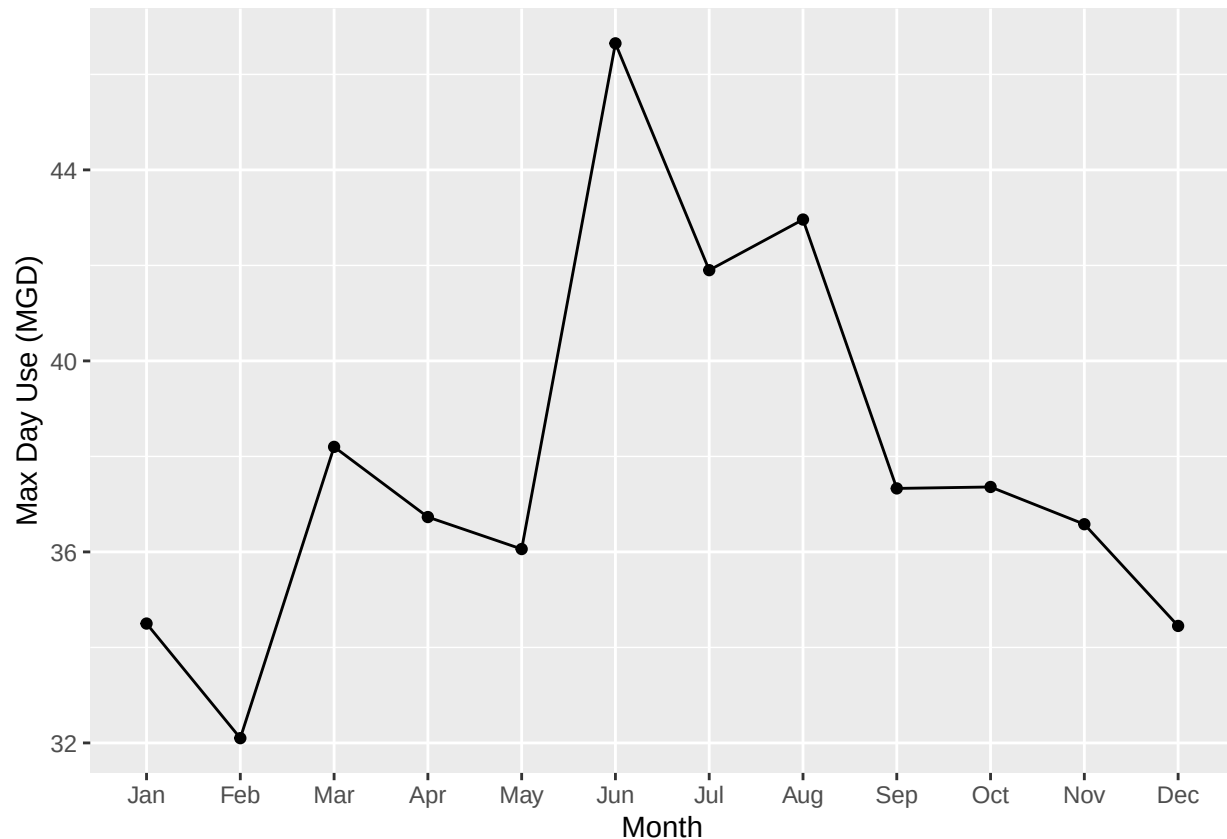
```
#4
Month <- month.abb
Year <- 2024

df_2024 <- data.frame(
  Month = Month,
  WaterSystemName = rep(water_system_name, 12),
  ContactPerson = rep(contact_person, 12),
  Ownership = rep(ownership, 12),
  MaxDayUse_MGD = as.numeric(max_day_use_mgd),
  Year = rep(Year, 12)
)
df_2024$Month <- factor(df_2024$Month, levels = month.abb)
df_2024$Date <- as.Date(paste(df_2024$Year, match(df_2024$Month, month.abb), 1, sep = "-"))
df_2024
```

```
##      Month WaterSystemName ContactPerson Ownership MaxDayUse_MGD Year
## 1     Jan           Durham Reginald Hicks Municipality      34.50 2024
## 2     Feb           Durham Reginald Hicks Municipality      32.10 2024
## 3     Mar           Durham Reginald Hicks Municipality      38.20 2024
## 4     Apr           Durham Reginald Hicks Municipality      36.73 2024
```

## 5	May	Durham Reginald Hicks Municipality	36.06	2024
## 6	Jun	Durham Reginald Hicks Municipality	46.65	2024
## 7	Jul	Durham Reginald Hicks Municipality	41.90	2024
## 8	Aug	Durham Reginald Hicks Municipality	42.96	2024
## 9	Sep	Durham Reginald Hicks Municipality	37.33	2024
## 10	Oct	Durham Reginald Hicks Municipality	37.36	2024
## 11	Nov	Durham Reginald Hicks Municipality	36.58	2024
## 12	Dec	Durham Reginald Hicks Municipality	34.45	2024
##	Date			
## 1	2024-01-01			
## 2	2024-02-01			
## 3	2024-03-01			
## 4	2024-04-01			
## 5	2024-05-01			
## 6	2024-06-01			
## 7	2024-07-01			
## 8	2024-08-01			
## 9	2024-09-01			
## 10	2024-10-01			
## 11	2024-11-01			
## 12	2024-12-01			

```
ggplot(df_2024, aes(x = Month, y = MaxDayUse_MGD, group = 1)) +
  geom_line() +
  geom_point() +
  labs(
    x = "Month",
    y = "Max Day Use (MGD)"
  )
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs, “PWSID” and “year”, that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

#6.

```
scrape_lwsp <- function(PWSID, year, MW_ID) {

  url <- paste0("https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=", PWSID, "&year=", year)
  website <- read_html(url)
  tables <- website %>% html_elements("table") %>% html_table(fill = TRUE)

  sys <- tables[[2]]
  water_system_name <- as.character(sys[1, 2])
  contact_person <- as.character(sys[4, 2])
  ownership <- as.character(sys[2, 5])

  mw <- as.matrix(tables[[MW_ID]])
  max_day_use_mgd <- as.character(c(mw[1:4, 3], mw[1:4, 6], mw[1:4, 9]))
}
```

```

df <- data.frame(
  PWSID = rep(PWSID, 12),
  Year = rep(year, 12),
  Month = month.abb,
  WaterSystemName = rep(water_system_name, 12),
  ContactPerson = rep(contact_person, 12),
  Ownership = rep(ownership, 12),
  MaxDayUse_MGD = as.numeric(max_day_use_mgd)
)

df$Date <- as.Date(paste(df$Year, match(df$Month, month.abb), 1, sep = "-"))
return(df)
}

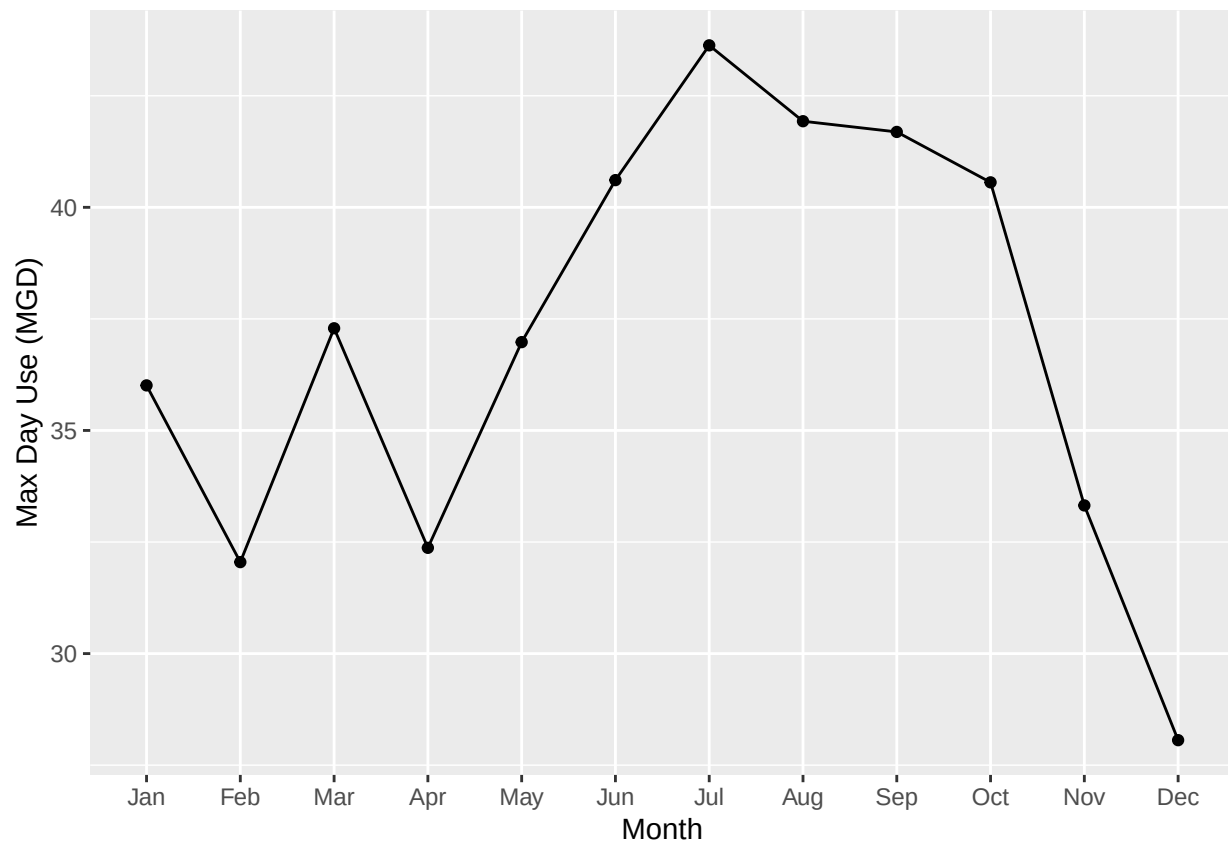
```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

#7
df_durham_2020 <- scrape_lwsp("03-32-010", 2020, 13)
df_durham_2020$Month <- factor(df_durham_2020$Month, levels = month.abb)
ggplot(df_durham_2020, aes(x = Month, y = MaxDayUse_MGD, group = 1)) +
  geom_line() +
  geom_point() +
  labs(x = "Month", y = "Max Day Use (MGD)")

```



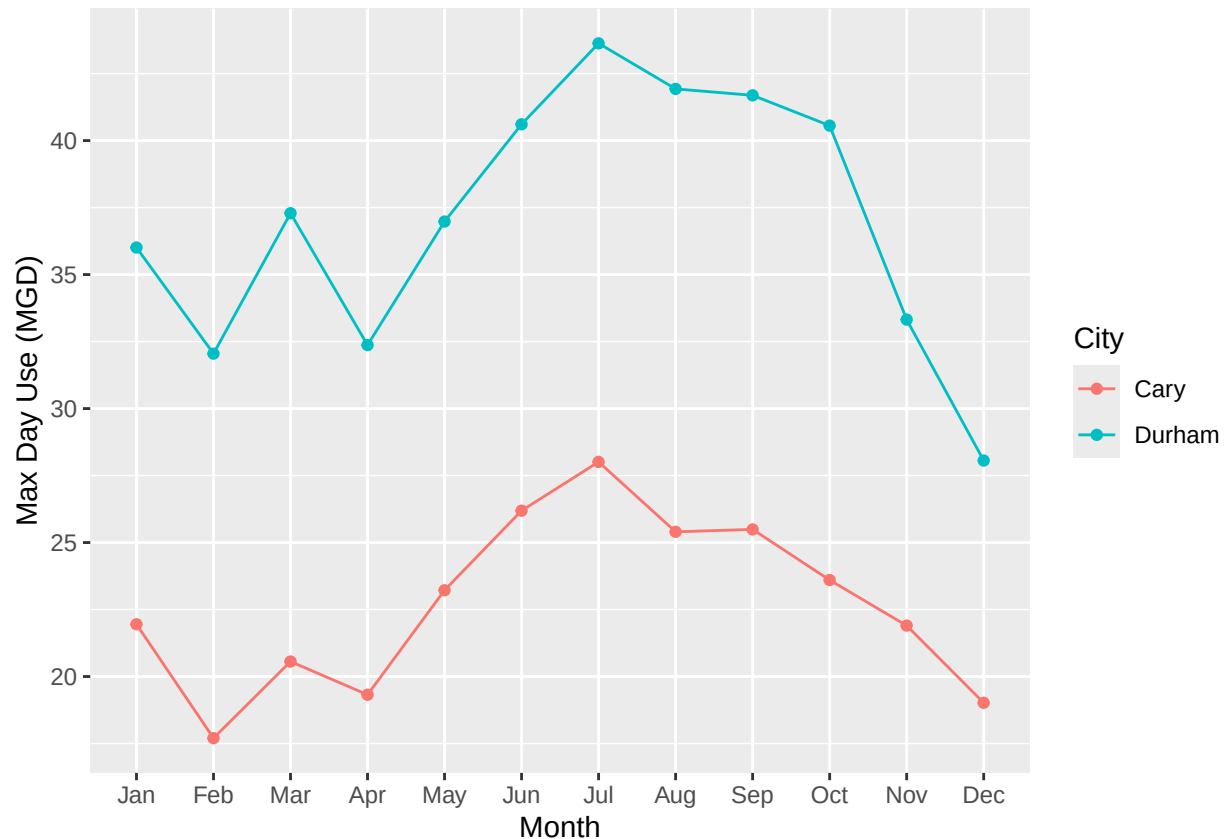
```
str(df_durham_2020)
```

```
## 'data.frame': 12 obs. of 8 variables:
## $ PWSID : chr "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
## $ Year : num 2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ Month : Factor w/ 12 levels "Jan","Feb","Mar",...: 1 2 3 4 5 6 7 8 9 10 ...
## $ WaterSystemName: chr "Durham" "Durham" "Durham" "Durham" ...
## $ ContactPerson : chr "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" "Reginald Hicks" ...
## $ Ownership : chr "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ MaxDayUse_MGD : num 36 32 37.3 32.4 37 ...
## $ Date : Date, format: "2020-01-01" "2020-02-01" ...
```

8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8
df_cary_2020 <- scrape_lwsp("03-92-020", 2020, 13)
df_both_2020 <- rbind(df_durham_2020, df_cary_2020)
df_both_2020$City <- ifelse(df_both_2020$PWSID == "03-32-010", "Durham", "Cary")
df_both_2020$Month <- factor(df_both_2020$Month, levels = month.abb)

ggplot(df_both_2020, aes(x = Month, y = MaxDayUse_MGD, color = City, group = City)) +
  geom_line() +
  geom_point() +
  labs(x = "Month", y = "Max Day Use (MGD)")
```



9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

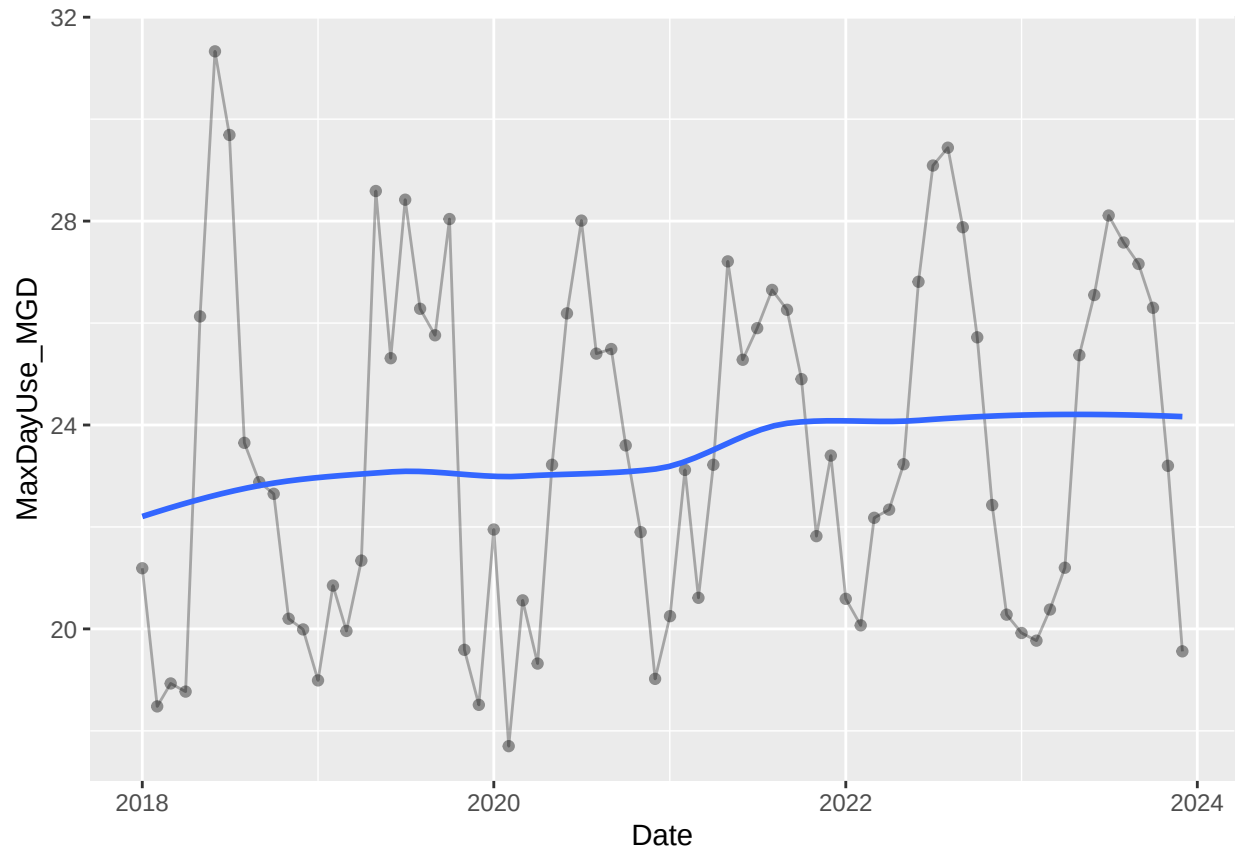
TIP: See Section 3.2 in the "06_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs (where one of the two inputs does not change). Pipe the output of the map2() function to `bind_rows()` to combine the dataframes into a single one, and use that to construct your plot.

```
#9
years <- 2018:2023
all_data <- NULL

for (y in years) {
  one_year <- scrape_lwsp("03-92-020", y, 13)
  all_data <- rbind(all_data, one_year)
}

ggplot(all_data, aes(Date, MaxDayUse_MGD)) +
  geom_line(alpha = 0.3) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "loess", se = FALSE)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: >Cary appears to have a slight upward trend in water usage over time. The blue smooth line gradually rises from around 23 MGD in 2018 to about 24 MGD by 2021–2023 and then levels off. Although there is strong seasonal variability each year, the overall baseline seems to increase slightly.